

DAY 2

Financial Statement Forecasting and Analysis

FMVA Certification Prep: Financial Modeling and Valuation Bootcamp

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Forecasting financial statements is the backbone of every valuation, credit decision, and strategic plan a finance professional builds. In this chapter you will move from raw historical data to a fully integrated three-statement model, apply ratio analysis to diagnose performance, and stress-test assumptions through scenario planning — giving you everything needed to answer FMVA exam questions on financial statement forecasting without consulting any other source.

Building a Revenue Forecast from Historical Trends and Drivers

Revenue forecasting is one of the most consequential steps in any financial model. Every downstream projection—gross profit, operating income, free cash flow, and ultimately valuation—inherits the assumptions you embed in the top line. A disciplined, driver-based approach forces you to defend each assumption explicitly and makes your model far more useful for scenario analysis.

Decomposing Revenue into Price × Volume

The most common forecasting error is treating revenue as a single, monolithic number and applying one growth rate to it. Instead, begin by separating revenue into its two fundamental components:

$$\text{Revenue} = \text{Units Sold} \times \text{Average Selling Price (ASP)}$$

This decomposition matters because price and volume respond to completely different forces. Volume is driven by market demand, distribution reach, new product launches, and competitive share shifts. ASP is driven by pricing power, product mix, discounting behavior, and inflation pass-through. When you blend them together, you can easily miss a situation where volume is growing 15% while ASP is eroding 5%—a very different business than one growing 10% uniformly.

For each segment, collect at least three years of unit shipments and realized ASPs separately. Even if the company does not disclose units directly, proxy KPIs—store count, active subscribers, square footage, or market share—can anchor the volume assumption to something observable and testable.

Establishing a Baseline with Historical CAGR

Once you have clean historical revenue data, calculate the compound annual growth rate over a three-to-five year window:

$$\text{CAGR} = (\text{Revenue_End} / \text{Revenue_Start})^{(1 / n)} - 1$$

This smooths out single-year noise and gives you a defensible starting point. Avoid anchoring on only one or two years; a single exceptional year can distort the picture dramatically.

Fully Worked Example

Suppose a company reported the following revenues:

- FY2020: \$800M
- FY2021: \$880M
- FY2022: \$968M

Step 1 — Calculate the historical CAGR over two years (FY20 to FY22):

$$\text{CAGR} = (\$968\text{M} / \$800\text{M})^{(1/2)} - 1 = (1.21)^{0.5} - 1 = 1.10 - 1 = 10.0\%$$

The implied two-year CAGR is exactly 10%.

Step 2 — Adjust for cycle and guidance (taper to 8%):

Management has guided toward moderating growth as the addressable market matures, and industry data suggests a mid-cycle slowdown. You apply an 8% growth rate for the forecast period—a reasonable taper from the 10% historical baseline.

Step 3 — Project forward using the core formula:

$$\text{Revenue}_t = \text{Revenue}_{\{t-1\}} \times (1 + g)$$

- FY2023: $\$968\text{M} \times 1.08 = \$1,045\text{M}$
- FY2024: $\$1,045\text{M} \times 1.08 = \$1,129\text{M}$
- FY2025: $\$1,129\text{M} \times 1.08 = \$1,219\text{M}$

Each year's projection flows mechanically from the prior year, making the model transparent and auditable.

Segment-Level Forecasting Reduces Error

Rather than applying 8% to total company revenue, best practice is to build the forecast segment by segment. A consumer electronics company might have a hardware segment growing at 5% and a services segment growing at 20%. Blending those into a single rate obscures the mix shift and produces a less accurate aggregate. Segment-level models also make it easier to incorporate company guidance, which is almost always given at the segment level, and to stress-test individual business units independently.

Anchoring to Observable KPIs

Every growth assumption should trace back to at least one observable KPI. If you are projecting 8% revenue growth, show that it implies, for example, 5% unit growth and 3% ASP improvement. Then ask: Is 5% unit growth consistent with the company's market share trajectory and the industry's projected unit demand? Is 3% ASP growth consistent with the company's pricing history and competitive dynamics? If the implied KPIs look unrealistic, revise the growth rate until the underlying drivers are coherent.

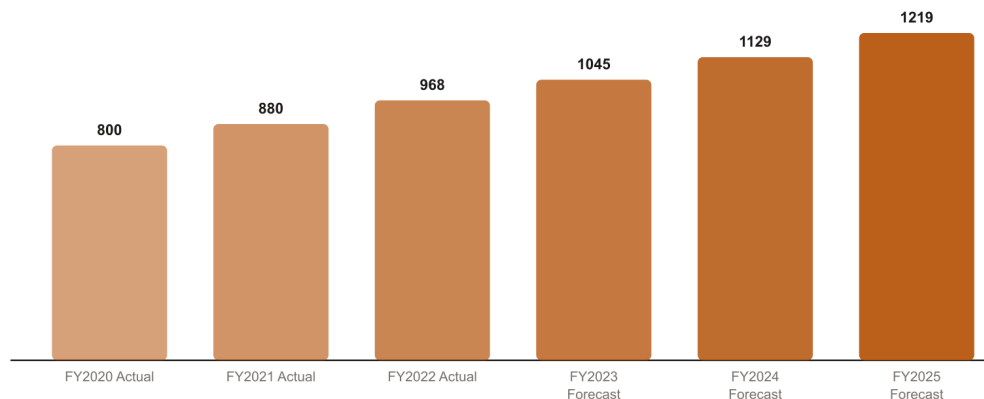
Documenting Assumptions for Sensitivity Analysis

Every assumption—historical CAGR, taper rate, segment growth rates, ASP trajectory—belongs in a dedicated input tab in your model, clearly labeled and linked to the forecast outputs. This single practice transforms a static projection into a living analytical tool. You can then build a sensitivity table that shows how FY2025 revenue changes if growth is 6%, 8%, or 10%, giving stakeholders an honest picture of forecast uncertainty.

Common mistake: Anchoring the forecast exclusively to the most recent year's growth rate rather than a multi-year CAGR. A single strong or weak year creates a biased baseline. Always use at least three years of history, and explicitly note any years distorted by one-time events before including them in the average.

Forecasting the Income Statement: COGS, Operating Expenses, and EBITDA Margins

Revenue Forecast: Historical & Projected



Historical 10% CAGR tapered to 8% reflecting market maturity and management guidance

Historical 10% CAGR tapered to 8% reflecting market maturity and management guidance

Projecting each line of the income statement with discipline and consistency is what separates a credible financial model from a rough estimate. The goal is to anchor every line item to a logical driver—revenue, an inflation index, a supporting schedule—rather than plugging in arbitrary growth rates.

Modeling COGS and Gross Margin

Cost of goods sold is almost always modeled as a percentage of revenue, because it is predominantly variable in nature. The standard approach is to calculate the gross margin % from three to five years of historical data, note any trend, and then blend the historical average with management's stated targets or industry benchmarks.

$$\text{Gross Profit} = \text{Revenue} \times \text{Gross Margin \%}$$

If a company has been running at 40–44% gross margin and management guides toward 42% through pricing discipline and procurement savings, anchoring your forecast at 42% is defensible and auditable.

Separating Fixed and Variable Operating Expenses

Operating expenses below gross profit—typically SG&A, R&D, and other overhead—require a split between fixed and variable components.

- Variable OpEx (commissions, shipping, certain marketing spend) scales directly with revenue, so model it as a percentage of revenue.
- Fixed OpEx (rent, base salaries, insurance) grows with inflation—typically 2–3% per year—regardless of revenue changes.

This distinction matters enormously in scenario analysis: a revenue shortfall compresses variable costs automatically but leaves fixed costs unchanged, correctly widening the loss.

D&A, Interest, and Tax: Link to Supporting Schedules

Three line items should never be forecasted in isolation:

- Depreciation & Amortization flows from the PP&E schedule. You build the schedule by adding capital expenditures, applying asset-class depreciation rates, and netting disposals. The resulting D&A figure drops into the income statement. Forecasting D&A as a flat percentage of revenue severs this logical link and creates errors when capex changes.
- Interest expense is linked to the debt schedule on the balance sheet. Beginning-period debt balance $\times$ weighted-average interest rate = interest expense for the period. This keeps the three financial statements integrated.
- Tax expense = Pre-tax income $\times$ effective tax rate. Use the company's historical effective rate (typically found in the tax footnote) unless management or legislation provides explicit guidance to the contrary.

$$\text{Tax Expense} = \text{Pre-Tax Income} \times \text{Effective Tax Rate}$$

$$\text{Net Income} = (\text{EBIT} - \text{Interest Expense}) \times (1 - \text{Effective Tax Rate})$$

EBITDA as the Bridge to Cash Flow

EBITDA is not just a valuation metric—it is the starting point for the cash flow statement under the indirect method.

$$\text{EBITDA} = \text{Revenue} - \text{COGS} - \text{Cash Operating Expenses}$$

Because D&A is a non-cash charge, stripping it out of operating income gives you a clean proxy for operating cash generation before working capital movements, capex, and financing costs.

Fully Worked Numerical Example

Assumptions: Revenue = \$1,045M | Gross margin = 42% | SG&A = 18% of revenue | D&A from PP&E schedule = \$55M | Interest expense from debt schedule = \$20M | Effective tax rate = 25%

Step 1 — Gross Profit

$$\text{Gross Profit} = \$1,045\text{M} \times 42\% = \$439\text{M}$$

Step 2 — SG&A

$$\text{SG\&A} = \$1,045\text{M} \times 18\% = \$188\text{M}$$

Step 3 — EBITDA

$$\text{EBITDA} = \$439\text{M} - \$188\text{M} = \$251\text{M}$$

EBITDA margin = $\$251\text{M} \div \$1,045\text{M} = 24.0\%$

Step 4 — EBIT (Operating Income)

$$\text{EBIT} = \text{EBITDA} - \text{D\&A} = \$251\text{M} - \$55\text{M} = \$196\text{M}$$

Step 5 — Pre-Tax Income

$$\text{Pre-Tax Income} = \text{EBIT} - \text{Interest} = \$196\text{M} - \$20\text{M} = \$176\text{M}$$

Step 6 — Tax Expense

$$\text{Tax} = \$176\text{M} \times 25\% = \$44\text{M}$$

Step 7 — Net Income

$$\text{Net Income} = \$176\text{M} - \$44\text{M} = \$132\text{M}$$

Net margin = $\$132\text{M} \div \$1,045\text{M} = 12.6\%$

Step-by-Step Forecasting Procedure

- Set revenue forecast using a driver-based or growth-rate method.
- Apply historical/target gross margin % to derive COGS and Gross Profit.
- Split OpEx into fixed (inflate by CPI) and variable (% of revenue); sum to total Cash OpEx.
- Calculate EBITDA = Gross Profit - Cash OpEx.
- Pull D&A from the completed PP&E schedule; subtract to get EBIT.
- Pull interest expense from the debt schedule; subtract to get Pre-Tax Income.
- Multiply Pre-Tax Income by the effective tax rate to get Tax Expense.
- Net Income = Pre-Tax Income - Tax Expense.

Common mistake: Forecasting D&A and interest expense as simple percentages of revenue rather than linking them to the PP&E and debt schedules. This breaks the integrity of the three-statement model—capex decisions no longer affect depreciation, and new borrowings no longer affect interest cost. Always build the supporting schedules first, then let those outputs feed the income statement.

Projecting the Balance Sheet: Working Capital, PP&E, and Debt Schedules

Forecasting the balance sheet is where a financial model either holds together or falls apart. Unlike the income statement, which flows naturally from revenue assumptions, the balance sheet requires deliberate, structured linkages — every line must be anchored to either a ratio-driven estimate or a supporting schedule. Master these mechanics and you will always produce a model that balances cleanly and tells a coherent story about a company's financial position.

Working Capital Items: Days-Based Ratio Forecasting

Accounts receivable (AR), inventory, and accounts payable (AP) are the three core working capital accounts, and all three are forecast using days-based ratios that translate operational efficiency assumptions into dollar balances.

The key ratios are:

- Days Sales Outstanding (DSO): measures how quickly customers pay
- Days Inventory Outstanding (DIO): measures how long inventory sits before sale
- Days Payable Outstanding (DPO): measures how long the company takes to pay suppliers

The conversion formulas are:

$$\text{AR} = \text{Revenue} \times (\text{DSO} / 365)$$

$$\text{Inventory} = \text{COGS} \times (\text{DIO} / 365)$$

$$\text{AP} = \text{COGS} \times (\text{DPO} / 365)$$

These ratios are typically derived from historical averages or industry benchmarks and held constant (or adjusted gradually) across the forecast period. Tying AR and inventory to revenue or COGS — rather than entering arbitrary dollar amounts — ensures the balance sheet responds realistically when your revenue assumptions change.

PP&E Roll-Forward

Property, plant, and equipment (PP&E) is forecast using a simple roll-forward schedule that tracks every dollar in and out of the account:

$$\text{Ending PP\&E} = \text{Beginning PP\&E} + \text{Capex} - \text{D\&A}$$

Capex is usually projected as a percentage of revenue or tied to a specific capital investment plan. Depreciation and amortization (D&A) flows directly from the income statement. This roll-forward keeps the income statement, cash flow statement, and balance sheet synchronized — a critical requirement for accounting integrity.

Debt Schedule

Long-term debt is tracked through a dedicated schedule that records:

- Beginning balance — prior period ending balance
- New borrowings — debt raised during the period
- Repayments — scheduled principal payments or optional paydowns
- Ending balance — Beginning + Borrowings - Repayments

Interest expense on the income statement is then calculated from the average or beginning debt balance, completing the link between the debt schedule and the P&L.

Retained Earnings Roll-Forward

Equity is updated through the retained earnings roll-forward:

$$\text{Ending RE} = \text{Beginning RE} + \text{Net Income} - \text{Dividends}$$

Net income comes directly from the income statement. Dividends are either modeled explicitly or held at zero for companies that do not distribute earnings. This formula is the mechanical bridge between the income statement and the balance sheet.

Enforcing the Balance: The Plug

After projecting every line, the balance sheet must satisfy:

$$\text{Total Assets} = \text{Total Liabilities} + \text{Total Equity}$$

In practice, a small imbalance almost always appears after the first pass because cash and debt are interdependent — more debt means more cash, but more cash reduces borrowing needs. The standard solution is to designate a plug account: typically a revolving credit facility (revolver) on the liabilities side if the company needs external funding, or a cash balance on the assets side if the company generates surplus cash. The plug is solved iteratively (or with a circular reference in Excel) until the sheet balances exactly.

Fully Worked Numerical Example

Given assumptions for Year 1 forecast:

| Item | Value |

---|---

| Revenue | \$1,045M |

| DSO | 45 days |

| Beginning PP&E | \$400M |

| Capex | \$70M |

| D&A | \$55M |

Step 1 — Forecast Accounts Receivable:

$$AR = \text{Revenue} \times (\text{DSO} / 365)$$

$$AR = \$1,045M \times (45 / 365)$$

$$AR = \$1,045M \times 0.12329$$

$$AR = \$128.8M$$

Step 2 — Forecast Ending PP&E:

$$\text{Ending PP\&E} = \text{Beginning PP\&E} + \text{Capex} - \text{D\&A}$$

$$\text{Ending PP\&E} = \$400M + \$70M - \$55M$$

$$\text{Ending PP\&E} = \$415M$$

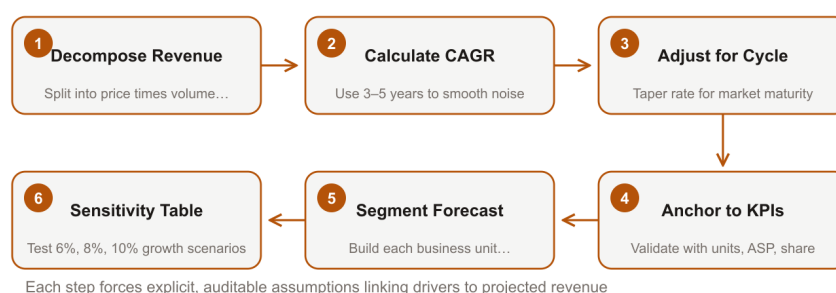
The \$15M net increase in PP&E (\$70M invested minus \$55M depreciated) reflects a company that is modestly expanding its asset base — a reasonable assumption to verify against management guidance or historical capex patterns.

Common mistake: Analysts sometimes project AR as a flat dollar amount carried forward from the prior year rather than recalculating it from the DSO formula. When revenue grows, this understates AR and artificially inflates operating cash flow — making the company look more cash-generative than it really is. Always recompute working capital balances from the ratio each period.

By chaining every balance sheet account to a ratio, a schedule, or a roll-forward, you create a model where changing a single revenue assumption ripples correctly through AR, inventory, AP, PP&E, retained earnings, and ultimately the plug — exactly the behavior a well-built forecast demands.

Linking the Three Statements into an Integrated Financial Model

Driver-Based Revenue Forecast Process



Each step forces explicit, auditable assumptions linking drivers to projected revenue

Building an integrated three-statement model is one of the most important skills in financial analysis. When the model is wired correctly, a single assumption change — say, a higher revenue growth rate — ripples automatically through the income statement, updates working capital on the balance sheet, adjusts operating cash flow, and recalculates the ending cash balance. The discipline of linking every line item consistently is what separates a professional model from a collection of disconnected schedules.

How Net Income Anchors the Cash Flow Statement

Under the indirect method, the cash flow statement begins with net income pulled directly from the bottom of the income statement. This single cell reference is the first link in the chain. From there, two categories of adjustments convert accrual-based net income into actual cash generated by operations.

Non-cash add-backs. Depreciation and amortization (D&A) was deducted on the income statement to match expenses with revenues, but no cash left the company. You therefore add it back in full.

Working capital adjustments. Changes in operating assets and liabilities capture the timing gap between recognizing revenue or expense and actually collecting or paying cash. The rule is straightforward: an increase in an operating asset (like accounts receivable) is a cash outflow because cash has not yet been collected, while an increase in an operating liability (like accounts payable) is a cash inflow because you have deferred a payment. The net change in working capital (9Dât2' —2 subtracted when it represents a net use of cash.

Capital Expenditures and the PP&E Roll-Forward

Capital expenditures appear as a cash outflow in the investing section of the cash flow statement. Simultaneously, they feed the PP&E roll-forward on the balance sheet:

$$\text{Ending PP\&E} = \text{Beginning PP\&E} + \text{Capex} - \text{Depreciation}$$

This roll-forward is a critical integrity check: if Capex in the cash flow statement does not match the Capex input driving the balance sheet, the model is broken. D&A flows in the opposite direction — it reduces PP&E on the balance sheet while being added back on the cash flow statement.

Debt Schedule and Financing Activities

A separate debt schedule tracks beginning balances, new borrowings, and repayments for each debt tranche. Net borrowings (new debt minus repayments) flow into the financing section of the cash flow statement and simultaneously update the long-term debt line on the balance sheet. Interest expense from the debt schedule feeds back into the income statement, which is where a circular reference can arise (discussed below).

The Integrity Check: Ending Cash

The single most important test of model integrity is:

$$\text{Ending Cash (Cash Flow Statement)} = \text{Cash (Balance Sheet)}$$

$$\text{Ending Cash} = \text{Beginning Cash} + \text{Operating Cash Flow} + \text{Investing Cash Flow} + \text{Financing Cash Flow}$$

If these two figures do not match to the penny, something is mis-linked. Analysts often color-code this check cell green or red to catch errors instantly.

Fully Worked Numerical Example

Assume the following for a single forecast year:

Item	Amount
Net Income	\$132M
D&A	\$55M
Increase in Net Working Capital (9Dât2' Â C „Ò À	
Capital Expenditures	\$70M
Net Borrowings	\$10M

| Beginning Cash | \$50M |

Step 1 — Operating cash flow:

Start with net income and adjust for non-cash items and working capital.

$$\text{Operating CF} = \$132\text{M} + \$55\text{M} - \$18\text{M} = \$169\text{M}$$

Step 2 — Investing cash flow:

Capex is a cash outflow.

$$\text{Investing CF} = -\$70\text{M}$$

Step 3 — Financing cash flow:

Net borrowings represent new debt raised in excess of repayments.

$$\text{Financing CF} = +\$10\text{M}$$

Step 4 — Change in cash:

$$\Delta \text{Cash} = \$169\text{M} - \$70\text{M} + \$10\text{M} = \$109\text{M}$$

Step 5 — Ending cash:

$$\text{Ending Cash} = \$50\text{M} + \$109\text{M} = \$159\text{M}$$

This \$159M must appear as the cash line on the balance sheet. If it does, the model balances.

The key profitability-to-cash bridge formula used in valuation contexts is:

$$\text{Free Cash Flow} = \text{EBIT} \times (1 - t) + \text{D\&A} - \text{Capex} - \Delta \text{WC}$$

Resolving Circular References

Interest expense depends on the average debt balance. Debt changes when cash is used as a plug to make the balance sheet balance. Cash depends on net income, which depends on interest expense. This loop creates a circular reference that Excel cannot resolve in a single pass. The professional solution is an iteration toggle: a Boolean switch cell that, when set to zero, breaks the circle by hardcoding interest to the prior period's value, and when set to one, allows Excel's iterative calculation setting to converge on a solution. Always document this toggle clearly so reviewers understand why iteration is enabled.

Common mistake: Forgetting to subtract the increase in working capital rather than add it. Analysts sometimes confuse the sign convention and inadvertently overstate operating cash flow — a model error that inflates free cash flow and, consequently, any DCF-based valuation built on top of it.

Financial Ratio Analysis: Diagnosing Performance and Benchmarking

Financial ratio analysis transforms raw financial statement numbers into meaningful signals about a company's health. A single dollar figure—say, \$132 million in net income—tells you almost nothing without context. Ratios create that context by expressing relationships between line items, and those relationships only become truly informative when compared against prior periods and industry peers.

Liquidity Ratios: Can the Business Meet Near-Term Obligations?

Liquidity ratios measure whether a company can cover its short-term liabilities with short-term assets.

$$\text{Current Ratio} = \text{Current Assets} / \text{Current Liabilities}$$

A current ratio above 1.0 suggests the firm has more liquid resources than near-term obligations. The quick ratio sharpens this test by removing inventory—which may be slow to convert to cash—from the numerator:

$$\text{Quick Ratio} = (\text{Current Assets} - \text{Inventory}) / \text{Current Liabilities}$$

A manufacturer with a current ratio of 2.1 but a quick ratio of 0.9 is signaling that most of its "liquidity" is tied up in inventory. That distinction matters enormously in a credit crunch.

Profitability Ratios: How Efficiently Is the Business Generating Earnings?

Moving down the income statement reveals progressively refined profitability:

- Gross Margin = Gross Profit / Revenue
- EBITDA Margin = EBITDA / Revenue
- Net Margin = Net Income / Revenue
- ROA = Net Income / Total Assets
- ROE = Net Income / Shareholders' Equity

Each margin strips away a different layer of cost. A company can show strong gross margins while poor net margins reveal excessive overhead or interest burden—a diagnostic pattern that guides management action.

Leverage Ratios: How Much Debt Risk Is the Business Carrying?

$$\text{Debt} / \text{EBITDA} = \text{Total Debt} / \text{EBITDA}$$

This is the dominant credit metric in leveraged finance; lenders typically grow concerned above 4–5×. Alongside it:

$$\text{Interest Coverage} = \text{EBIT} / \text{Interest Expense}$$

A ratio below 2× raises serious solvency questions. Both ratios should be trended over time—deterioration across two or three periods is more alarming than a single weak reading.

Efficiency Ratios and the Cash Conversion Cycle

$$\text{Asset Turnover} = \text{Revenue} / \text{Total Assets}$$

Higher turnover means the firm extracts more revenue per dollar of assets deployed. Working-capital efficiency is captured by three companion metrics that combine into the Cash Conversion Cycle (CCC):

- DSO (Days Sales Outstanding) = (Receivables / Revenue) × 365
- DIO (Days Inventory Outstanding) = (Inventory / COGS) × 365
- DPO (Days Payable Outstanding) = (Payables / COGS) × 365
- CCC = DSO + DIO - DPO

A shorter CCC means cash cycles back to the business faster, reducing the need for external financing.

DuPont Decomposition: Unpacking ROE

The most powerful diagnostic framework in ratio analysis is the three-factor DuPont identity:

$$\text{ROE} = (\text{Net Income} / \text{Revenue}) \times (\text{Revenue} / \text{Assets}) \times (\text{Assets} / \text{Equity})$$

This rewrites ROE as the product of Net Margin × Asset Turnover × Equity Multiplier, isolating whether returns are driven by profitability, operational efficiency, or financial leverage. Two companies can post identical ROEs through completely different strategies—and DuPont reveals which levers each is pulling.

Worked Example: DuPont Decomposition

Assume the following for Meridian Industrial Corp (fiscal year-end):

Item	Value
Net Income	\$132M
Revenue	\$1,050M
Total Assets	\$1,100M
Shareholders' Equity	\$660M

| Net Income | \$132M |

| Revenue | \$1,050M |

| Total Assets | \$1,100M |

| Shareholders' Equity | \$660M |

Step 1 — Net Margin:

$$\$132\text{M} \div \$1,050\text{M} = 12.6\%$$

Step 2 — Asset Turnover:

$$\$1,050\text{M} \div \$1,100\text{M} = 0.95\times$$

Step 3 — Equity Multiplier:

$$\$1,100\text{M} \div \$660\text{M} = 1.67\times$$

Step 4 — Reconstruct ROE:

$$0.126 \times 0.95 \times 1.67 = \sim 20.0\%$$

Direct check: $\$132\text{M} \div \$660\text{M} = 20.0\%$ '

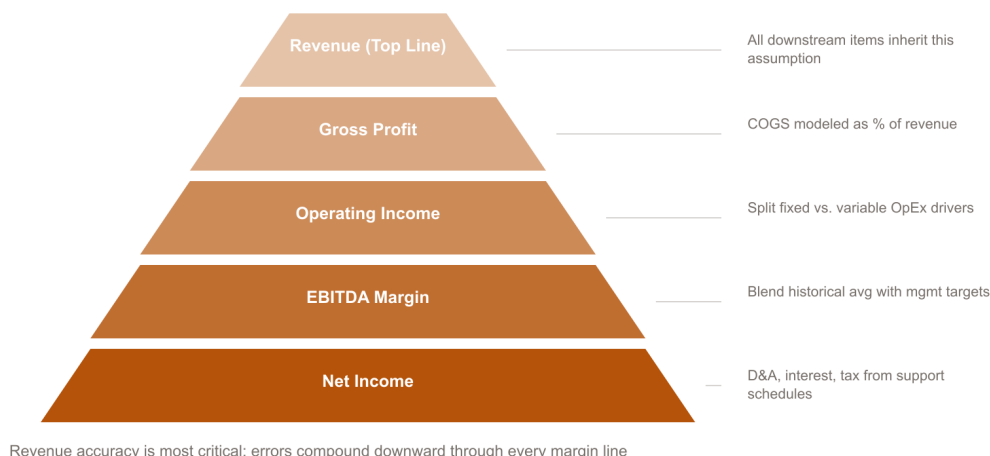
Meridian's 20% ROE is primarily driven by a solid net margin and modest leverage—not aggressive asset loading. If a peer also posts 20% ROE but with an equity multiplier of 3.0×, that peer is taking on substantially more financial risk to achieve the same headline return.

Common mistake: Analysts often compare a single ratio to a round-number benchmark ("a current ratio should be 2.0") without referencing the company's own history or its industry norm. Capital-light software businesses routinely run current ratios below 1.0 and remain perfectly healthy; capital-intensive manufacturers may need 2.5× to feel comfortable. Always anchor your interpretation to a relevant comparison set—prior-year trend, sector median, or a direct competitor—before drawing conclusions.

Ratio analysis is not a checklist to complete; it is a diagnostic conversation with the financial statements. Each ratio raises a question, and the answer usually requires examining two or three others alongside it.

Scenario and Sensitivity Analysis: Stress-Testing the Forecast

Income Statement Forecasting Hierarchy



Revenue accuracy is most critical; errors compound downward through every margin line

Stress-testing a forecast transforms a single-point projection into a decision-useful range of outcomes. Rather than defending one set of assumptions, you present stakeholders with a structured view of upside, base, and downside—and, critically, you show which levers matter most.

Identifying the High-Impact Assumptions

Before building any table, rank your assumptions by their leverage on the output metric. For most operating companies, revenue growth rate and EBITDA margin account for the majority of variance in Net Income and Debt/EBITDA. A 100-basis-point (bps) shift in revenue growth on a \$1 billion revenue base moves top-line by \$10 million; the same 100 bps shift in EBITDA margin moves EBITDA by \$10 million directly—no volume effect required. Cost structure assumptions (D&A, interest rate on debt) and tax rate matter, but typically rank third and fourth in a tornado chart.

Sensitivity = $\Delta \text{EBITDA} / (\text{Revenue Growth Rate} \times \text{Revenue Base})$, expressed as \$ change per 100 bps

For a company with \$1 B base revenue and 20% EBITDA margin, a 100 bps increase in revenue growth adds roughly \$2 M–\$3 M to EBITDA after accounting for variable cost flow-through—far less than a 100 bps margin improvement, which adds \$10 M directly. This asymmetry is the core insight the tornado chart will visualize.

Building the Scenario Toggle

Create a named range called Scenario driven by a data-validation dropdown containing "Bear," "Base," and "Bull." Use INDEX/MATCH to pull every linked input from a scenario table:

```
=INDEX(ScenarioTable, MATCH($B$2, ScenarioNames, 0), MATCH(InputName, InputHeaders, 0))
```

This single toggle simultaneously switches revenue growth, EBITDA margin, capex rate, and working capital assumptions. Analysts cannot accidentally mix a bull revenue assumption with a bear margin—a common and dangerous error.

Common mistake: Building three separate hard-coded columns instead of a toggle causes version drift. When you update the depreciation schedule in the base case, the bear and bull cases silently become stale. Always drive all scenarios from one formula-linked table.

One-Variable Sensitivity: Revenue Growth vs. EBITDA

Vary revenue growth from " 5% to +15% in 5% steps, holding margin constant at the base case 20%. On a \$1 B base:

Revenue Growth	Revenue (\$M)	EBITDA (\$M)
" 5%	950	190
0%	1,000	200
+5%	1,050	210
+10%	1,100	220
+15%	1,150	230

Each 500 bps step adds \$10 M to EBITDA—a clean linear relationship confirming the sensitivity formula above.

Two-Variable Data Table: The 3×3 Grid

Now cross revenue growth (6%, 8%, 10%) with EBITDA margin (18%, 20%, 22%) to produce Net Income. Assume: base revenue \$1 B, D&A \$40 M, interest expense \$30 M, tax rate 25%.

Step 1 — Calculate EBIT: $EBIT = EBITDA - D\&A$

Step 2 — Calculate EBT: $EBT = EBIT - \text{Interest}$

Step 3 — Calculate Net Income: $\text{Net Income} = EBT \times (1 - \text{Tax Rate})$

Fully worked cell: Revenue growth 8%, EBITDA margin 20%

- Revenue = $\$1,000 \text{ M} \times 1.08 = \$1,080 \text{ M}$
- EBITDA = $\$1,080 \text{ M} \times 20\% = \216 M
- EBIT = $\$216 \text{ M} - \$40 \text{ M} = \$176 \text{ M}$
- EBT = $\$176 \text{ M} - \$30 \text{ M} = \$146 \text{ M}$
- Net Income = $\$146 \text{ M} \times 0.75 = \$109.5 \text{ M} \approx \110 M

Completing all nine cells:

| | Margin 18% | Margin 20% | Margin 22% |

|---|---|---|---|

| Growth 6% | \$98 M | \$121 M | \$143 M |

| Growth 8% | \$105 M | \$110 M* | \$155 M |

| Growth 10% | \$113 M | \$138 M | \$168 M |

Rounded for illustration; full precision shown in worked step above.

Reading across any row, a 200 bps margin improvement adds roughly \$22 M–\$28 M to Net Income. Reading down any column, a 400 bps growth improvement adds roughly \$15 M–\$25 M. Margin is the dominant driver—its column-to-column swing exceeds the row-to-row swing from growth, confirming the tornado chart ranking.

Tornado Chart and Goal-Seek

Rank each assumption by its total swing (high case minus low case output) and plot horizontal bars from largest to smallest. Margin appears at the top; revenue growth second; interest rate and tax rate trail. This visualization immediately communicates where management attention—and negotiation leverage—should focus.

Finally, flip the logic: instead of asking "what does Net Income equal given these assumptions," use goal-seek to ask "what EBITDA margin is required to achieve \$130 M Net Income at 8% growth?" The answer (~21.1%) becomes a concrete operational target, bridging forecast analysis directly into strategic planning.

| Glossary

- CAGR — Compound annual growth rate; smoothed annual return over multiple periods.
- DSO — Days Sales Outstanding; average days to collect accounts receivable.
- DIO — Days Inventory Outstanding; average days inventory is held before sale.
- DPO — Days Payable Outstanding; average days taken to pay suppliers.
- EBITDA — Earnings before interest, taxes, depreciation, and amortization; proxy for operating cash.

- Free Cash Flow — Cash generated after operating costs and capital expenditures; available to investors.
- Debt/EBITDA — Leverage ratio measuring years of earnings needed to repay total debt.
- DuPont Analysis — ROE decomposition into margin, asset turnover, and financial leverage components.

